

# NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

## Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

May - June 2023

20EC603 / 17EC502 / 16EC502 / 15EC601 – INFORMATION THEORY AND CODING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

### Unit – I

Marks BT\* CO\* PO\*

1. a) Suppose that  $X$  is a random variable whose entropy  $H(X)$  is 8 bits/sym. Suppose that  $Y(X)$  is a deterministic function that takes on a different value for each value of  $X$ .
  - (i) Determine the entropy of  $Y$ ?
  - (ii) Consider that the deterministic function  $Y(X)$  is not invertible; in other words, different values of  $X$  may correspond to the same value of  $Y(X)$ . In that case, what inference can be made about change in entropy,  $H(Y)$  determined in (i)?
- b) Determine the relationship between nat and a quadraple. (Quadraple is the unit of information associated with base 4).
- c) Consider a Markoff source as shown in Fig. Q 1(c). Determine
  - i. State Probabilities
  - ii. State Entropies
  - iii. Source Entropy

6 L\*4 1 2

5 L3 1 1

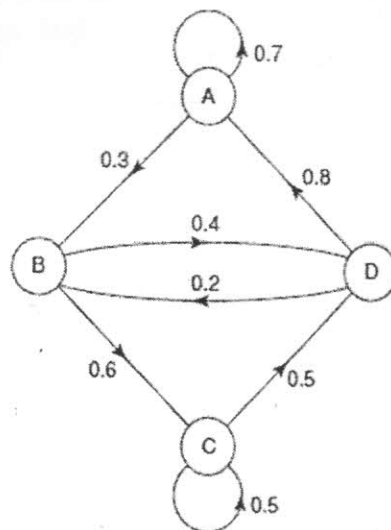


Fig. Q 1(c)

9 L3 1 1

2. a) Several people at a meeting are trying to guess a 3-bit binary number.
  - i. Person P is told that the number is odd
  - ii. Person Q is told that the number is not a multiple of 3 (i.e., not 0, 3, or 6)
  - iii. Person R is told that the number contains exactly two 1's
  - iv. And Person S is given all three of these clues.

How much information (in bits) did each person get about the number?

10 L4 1 2

- b) Consider a Markoff source as shown in Fig. Q 2(b). Determine
- State Probabilities.
  - State Entropies.
  - Source Entropy.
  - $G_1$  and  $G_2$ . Show that  $G_1 > G_2 > H$ .

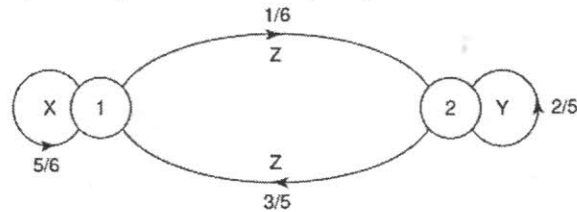


Fig. Q 2(b)

**Unit – II**

3. a) What is the shortest possible average codeword length, in bits per symbol, that could be achieved for a six-letter alphabet whose symbols have the following probability distribution? {0.5, 0.25, 0.125, 0.0625, 0.03125, 0.03125}.
- b) Consider the following source  
 $S = \{A, B, C, D, E, F\}$   
 $P = \{0.5, 0.2, 0.1, 0.1, 0.08, 0.02\}$   
 Find the codewords using Shannon Fano algorithm. Also find the coding efficiency and redundancy.
- c) Encode the following information using LZ algorithm.  
 sir\_sid\_eastman\_easily\_teases\_sea\_sick\_seal.
4. a) The following coding scheme has been used to encode the sequence.

| Source Symbols | Code B |
|----------------|--------|
| I              | 1      |
| S              | 01     |
| H              | 001    |
| T              | 0001   |
| A              | 00001  |
| -              | 00000  |

Draw the decision tree.

If the receiver receives the following bit-stream, decode the information.

00010011010000010100000001101000000010000100010000000  
 00100110100000001 000010001000001010000000110100000

- b) A source emits one of the eight symbols with the following probabilities  
 $P = (0.30, 0.25, 0.22, 0.07, 0.06, 0.05, 0.03, 0.02)$ .  
 Find the binary and ternary codewords using Huffman algorithm.  
 Find the efficiency in each case.
- c) Consider a discrete memoryless source with  $S = (X, Y, Z)$  with respective probabilities  $P = (0.5, 0.3, 0.2)$ . Find the codeword for the message 'YYZZX' using arithmetic coding.

10 L3 1 1

4 L3 2 1

7 L3 2 1

9 L3 2 1

5 L3 2 1

8 L3 2 1

7 L3 2 1

## Unit – III

5. a) Channel matrix for a channel is given as follows:

$$P\left(\frac{Y}{X}\right) = \begin{matrix} & y_1 & y_2 & y_3 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.3 & 0.2 & 0.5 \\ 0.1 & 0.4 & 0.5 \end{bmatrix} \end{matrix}$$

Also, the input probabilities of the channel are

$$P[X] = \left[ \frac{1}{2}, \frac{1}{4}, \frac{1}{4} \right]$$

Find

- i.  $H(X)$
- ii.  $H(Y)$
- iii.  $H(X, Y)$
- iv.  $H\left(\frac{Y}{X}\right)$
- v.  $H\left(\frac{X}{Y}\right)$

10 L3 3 1

- b) Derive the expression for the channel capacity of a symmetric channel with 'M' outputs. Hence find the channel capacity for a binary symmetric channel with error probability being 0.2.

10 L3 3 1

6. a) For the JPM given, find

- i.  $H(X)$
- ii.  $H(Y)$
- iii.  $H(X, Y)$
- iv.  $H(X/Y)$
- v.  $H(Y/X)$

$$\text{JPM} = P(X, Y) = \begin{matrix} & y_1 & y_2 & y_3 & y_4 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 0.05 & 0 & 0.20 & 0.05 \\ 0 & 0.10 & 0.10 & 0 \\ 0 & 0 & 0.20 & 0.10 \\ 0.05 & 0.05 & 0 & 0.10 \end{bmatrix} \end{matrix}$$

10 L3 3 1

- b) What are Binary Erasure Channels? Derive the expression to find the capacity of Binary Erasure Channels.

10 L2 3 1

## Unit – IV

7. a) Suppose that the management has decided to use a 20-bit data blocks in the company's new (n, 20) error correcting code. It is a single bit error correcting code. What is the minimum value of 'n' that will permit the code to be used for single bit error correction?
- b) Show that all error patterns that differ by a code vector have the same syndrome.
- c) The generator polynomial of a (15, 7) cyclic code is  $g(x) = 1 + x^4 + x^6 + x^7 + x^8$ . Find the codeword in systematic form for the message (1 0 1 1 1 0 1), clearly showing the states of registers. Verify your answer by direct computation.

5 L4 4 2

5 L2 4 1

10 L3 4 1

8. a) A communication involving three data bits uses a systematic block code with the following parity equations:

$$P1 = D1 + D2$$

$$P2 = D2 + D3$$

$$P3 = D3 + D1$$

Where, (D1, D2, D3) are data and (P1, P2, P3) are parity bits.

- What are the values of n and k for this code?
  - Find the minimum Hamming distance of this code?
- b) Find the set of 16 binary linear block codewords of length 7 such that each codeword is at a distance of atleast 3 from any other codeword.
- c) For a (7, 4) cyclic code with  $g(x) = 1 + x + x^3$ , draw the encoder diagram. Hence obtain the encoded output for the input  $m(x) = 1 + x^2$ .

6 L3 4 1

8 L4 4 2

6 L3 4 1

### Unit – V

9. a) Consider two non-systematic convolutional coding schemes - I and II. The generator polynomials for the two schemes are

Scheme I:  $G0 = 1101$ ,  $G1 = 1110$

Scheme II:  $G0 = 110101$ ,  $G1 = 111011$

- What are the code rates of schemes I and II?
- Find the number of states in the state diagram representations of Schemes I and II?
- Which code will lead to a lower bit error rate? Why?
- Write the encoder diagrams for Schemes I and II.

10 L4 5 2

- b) Consider a (2, 1, 2) convolution code with the generator sequences  $g_1 = 101$ ,  $g_2 = 111$ . Consider a received sequence (10, 01, 11, 11, 11, 10, 11).

Decode the information sequence using hard decision Viterbi algorithm.

10 L3 5 1

10. a) Consider a non-systematic (3, 1, 2) convolution code which has the following generator sequences:

$g_1 = 110$ ,  $g_2 = 101$  and  $g_3 = 111$

- Draw the encoder diagram.
- Obtain its generator matrix.
- Obtain the codeword for the information sequence (101001) in time domain approach.
- Verify the code obtained in (iii) using transform domain approach.

10 L3 5 1

- b) Given a (2, 1, 2) binary convolution code with the partially completed state diagram as shown in Fig. Q 10(b). Construct the complete state diagram and sketch the encoder diagram. Dotted lines correspond to input 1 and thick lines correspond to input 0.

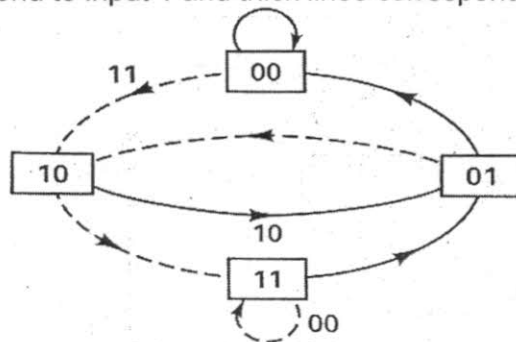


Fig. Q 10(b)

10 L4 5 2

BT\* Bloom's Taxonomy, L\* Level;

CO\* Course Outcome; PO\* Program Outcome

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